

Department of Computer Science and Engineering

Course Code: CSE370	Credits: 3.0
Course Name: Database Systems	Semester: Spring 2026

Lab 02: SQL Subqueries & Aggregate Functions

Activity List

- All commands are shown in the red boxes.
- In the green box, write the appropriate query/answer.
- All new queries should be typed in the command window after `mysql>`
- Start by connecting to the server using: `mysql -u root -p [password: <just press enter>]`
- For more MySQL queries, go to www.w3schools.com/sql or google it!

Initial Table: It's a bit different than Lab 01!

std_id	name	major	section	days_present	project_marks	cgpa	submission_date
s001	Abir	CS	1	10	18.5	3.91	2018-09-15
s002	Nafis	CSE	1	12	20	3.86	2018-08-15
s003	Tasneem	CS	1	8	18	3.57	2018-09-18
s004	Nahid	ECE	2	7	16.5	3.25	2018-08-20
s005	Arafat	CS	2	11	20	4.0	2018-09-13
s006	Tasneem	CSE	1	12	17.5	3.7	2018-08-15
s007	Muhtadi	ECE	1	10	19	3.67	2018-09-16

Link for Table Data: https://docs.google.com/document/d/1j6zmKf3cUr7zOKNxByfRPh0_AaTUn3hTyW6_AKuCFQA/

The purpose of the SELECT statement is to retrieve and display data from one or more database tables. It is an extremely powerful command. SELECT is the most frequently used SQL command and has the following general form:

SELECT [DISTINCT | ALL] { * | [columnExpression [AS newName]] [, . . .] }

FROM TableName [alias] [, . . .]

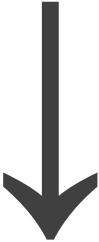
[WHERE condition]

[GROUP BY columnList] [HAVING condition]

[ORDER BY columnList]

columnExpression represents a column name or an expression, TableName is the name of an existing database table or view that you have access to, and alias is an optional abbreviation for TableName.

The sequence of processing in a **SELECT** statement is:

	FROM specifies the table or tables to be used
	WHERE filters the rows subject to some condition
	GROUP BY forms groups of rows with the same column value
	HAVING filters the groups subject to some condition
	SELECT specifies which columns are to appear in the output
	ORDER BY specifies the order of the output

The order of the clauses in the **SELECT** statement cannot be changed. The only two mandatory clauses are the first two: **SELECT** and **FROM**; the remainder are optional. The **SELECT** operation is closed: the result of a query on a table is another table.

Task 1: Aggregate Functions, Group By and Having:

- How will you retrieve the last submission date?

- What is the purpose of the group by keyword? In the above command, if we group by sub_date, instead of major, what will be the output?

- The having and where clauses are both used to specify a condition when selecting rows. What is the difference between them?

Task 2: Sub Queries/Nested Queries, Any and All:

- Think about how you can retrieve the names of students who got the highest project marks. Try out your query. Did you get the “correct” response according to the table?

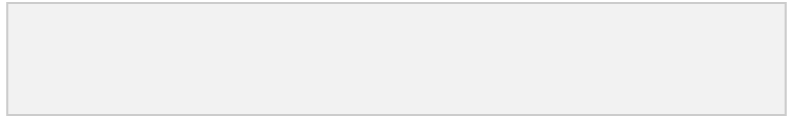
- Did you understand the role of “any” and “all” in the above queries? Explain below.

- Retrieve the names of the students who have received marks greater than at least 1 student doing the same major as them.[Hint: see next command]

Task 3: Correlated Subqueries and Exists:

- L1 and L2 are temporary aliases and create two separate instances for Lab_Grades; why are they required?

- Please identify the difference between the above two queries. [Hint: 1 asks for unique-only 1 student got the highest and the other didn't]



Task 4: Take a Quiz

Go to https://sqlzoo.net/wiki/Nested_SELECT_Quiz to test your understanding of the queries taught in class.